

① Sean $\sigma, T: \mathbb{N} \rightarrow \mathbb{N}$ estrictamente crecientes / $\sigma(N) \leq T(N)$.

Probar $\sigma(n) \geq T(n) \forall n \in \mathbb{N}$ y $\{x_{\sigma(n)}\}$ es parcial de $\{x_{T(n)}\}$

$\forall n \in \mathbb{N}, \exists m \in \mathbb{N} / \sigma(n) = T(m)$. ¿ $m \geq n$?

Se tiene $n > n-1 > \dots > 0 \Rightarrow \sigma(n) > \sigma(n-1) > \dots > \sigma(0)$

$T(m) > T(m-1) > \dots > T(m_0)$

$m > m-1 > \dots > m_0 \geq 0$

Si $m < n$, contradicción $\Rightarrow m \geq n \Rightarrow \sigma(n) = T(m) \geq T(n) \forall n \in \mathbb{N}$

Sea $S: \mathbb{N} \rightarrow \mathbb{N}$ estrictamente creciente y bien definida.
 $n \mapsto m_n$

$\sigma(n) = T(S(n)) \Rightarrow \{x_{\sigma(n)}\} = \{x_{T(S(n))}\}$

② Probar $\chi(t) = \sqrt{t} \quad t \in]0, +\infty[$ no puede ser solución
de $\mathbb{E}: \mathbb{R}^2 \rightarrow \mathbb{R} \quad (t, v) \mapsto \mathbb{E}(t, v)$ continua.

Sea (PC) $\dot{x} = \Sigma(t, x)$ $x(t_0) = x_0$, $\Sigma: D \subseteq \mathbb{R} \times \mathbb{R}^d \rightarrow \mathbb{R}^d$ continua, $(t_0, x_0) \in D$.

Probar $\exists$ sol. maximal.

Por Cauchy-Peano, $\exists I = I_0 \subseteq \mathbb{R} / x: I \rightarrow \mathbb{R}^d$ es sol. de (PC).

Utilizamos un razonamiento análogo al Teo Hahn-Banach.

Sea $\mathcal{L} = \{ \tilde{x}: \tilde{I} \rightarrow \mathbb{R}^d / \tilde{x} \text{ sol. de (PC)}, I \subseteq \tilde{I} \text{ abierto}, \tilde{x}(t) = x(t) \forall t \in I \}$

Definimos $\forall \tilde{x}_1, \tilde{x}_2 \in \mathcal{L}, x_1 \leq x_2 \iff \begin{cases} \tilde{I}_1 \subseteq \tilde{I}_2 \\ \tilde{x}_1(t) = \tilde{x}_2(t) \forall t \in \tilde{I}_1 \end{cases}$ veamos que es un orden.

1) $\tilde{x} \leq \tilde{x} \quad \forall \tilde{x} \in \mathcal{L}$

2) $\tilde{x}_1 \leq \tilde{x}_2, \tilde{x}_2 \leq \tilde{x}_1 \Rightarrow \begin{cases} \tilde{I}_1 = \tilde{I}_2 \\ \tilde{x}_1(t) = \tilde{x}_2(t) \forall t \in \tilde{I}_1 = \tilde{I}_2 \end{cases} \Rightarrow \tilde{x}_1 = \tilde{x}_2$

3) $\tilde{x}_1 \leq \tilde{x}_2, \tilde{x}_2 \leq \tilde{x}_3 \Rightarrow \tilde{I}_1 \subseteq \tilde{I}_2 \subseteq \tilde{I}_3, \tilde{x}_3(t) = \tilde{x}_1(t) \forall t \in \tilde{I}_1$

Sea $\mathcal{L} = \{ x_\gamma: I_\gamma \rightarrow \mathbb{R}^d / \gamma \in \mathcal{I} \} \subseteq \mathcal{L}$ totalmente ordenado.

Sea $I_* = \bigcup_{\gamma \in \mathcal{I}} I_\gamma$, que es claramente un intervalo abierto.

Sea $x_*: I_* \rightarrow \mathbb{R}^d / x_*(t) = x_\gamma(t)$ si $t \in I_\gamma$

Si $t \in I_* \Rightarrow \exists \gamma \in \mathcal{I} / t \in I_\gamma$. sea $\tilde{\gamma} \in \mathcal{I}, \tilde{t} \in I_{\tilde{\gamma}}$

Podemos suponer $x_{\tilde{\gamma}} \leq x_\gamma \Rightarrow I_{\tilde{\gamma}} \subseteq I_\gamma$ y $x_{\tilde{\gamma}}(t) = x_\gamma(\tilde{t}) \quad \forall \tilde{t} \in I_{\tilde{\gamma}}$

Sea $t_1 \in I_* \Rightarrow \exists \gamma \in \mathcal{I} / t_1 \in I_\gamma \Rightarrow x_*(t) = x_\gamma(t) \quad \forall t \in I_\gamma$
 $(x_*)|_{I_\gamma} = x_\gamma \Rightarrow (x_*)|_{I_\gamma}(t) = x_\gamma(t) = \Sigma(t, x_\gamma(t)) \quad \forall t \in I_\gamma \Rightarrow x_\gamma(t) = x_\gamma(t)$

$x_*(t) = x_\gamma(t)$ si $t \in I_\gamma$ y $x_*(t_0) = x_\gamma(t_0) = x_0$

Por lema de Zorn $\exists \bar{x}: \bar{J} \rightarrow \mathbb{R}^d$ elemento maximal de $\mathcal{L}$. Veamos que es solución maximal de (PC).

Por red. al absurdo, sup. $\bar{x}$ no es sol. maximal $\Rightarrow \exists \tilde{x}: \tilde{J} \rightarrow \mathbb{R}^d$ sol. de (PC)

con $\bar{J} \not\subseteq \tilde{J}$ abierto (lema prolongabilidad) y $\tilde{x}(t) = \bar{x}(t) \quad \forall t \in \bar{J} \Rightarrow \tilde{x} \in \mathcal{L}$ y $\tilde{x} > \bar{x} !!!$

Contradicción pues $\bar{x}$ era elemento maximal.

Consideramos $\mathbb{X}: \mathbb{R} \times \mathbb{R}^+ \rightarrow \mathbb{R}$ cont. y el (PC) $\begin{cases} \dot{x} = \mathbb{X}(t, x) \\ x(0) = x_0 > 0 \end{cases}$

Se supone $\exists m, \mu: \mathbb{R} \rightarrow \mathbb{R}^+ \quad |\mathbb{X}(t, x)| \leq m(t) \|x\| + \mu(t) \quad \forall (t, x) \in D$ y $\exists \mu, \varepsilon > 0 / \mathbb{X}(t, x) \geq \mu \quad \forall x \in [0, \varepsilon]$. Probar que la solución está definida a $+\infty$.

Sea $I =]t, w[$. Sup. $w < +\infty \Rightarrow$

1) $\lim_{t \rightarrow w^-} \|x(t)\| = +\infty \vee 2) \exists t_1, t_2 \nearrow w / x(t_1) \rightarrow \{ (w, \eta) \in \partial D$

- Veamos 2) no se cumple. $\partial D = \mathbb{R} \times \{0\} \Rightarrow \eta = 0$

$\Rightarrow \exists t^* \in [0, w[/ x(t^*) = \varepsilon, \dot{x}(t^*) \leq 0$

pero $\dot{x}(t^*) = \mathbb{X}(t^*, x(t^*)) = \mathbb{X}(t^*, \varepsilon) \geq \mu > 0$!!!

- Veamos 1) tampoco se cumple.

$x(t) = x_0 + \int_0^t \mathbb{X}(s, x(s)) ds$. Sea $M = \max_{t \in [0, w]} m(t)$, $N = \max_{t \in [0, w]} \mu(t)$

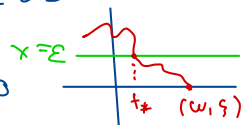
$$|x(t)| \leq |x_0| + \int_0^t m(s) |x(s)| + \mu(s) ds \leq |x_0| + Nt + M \int_0^t |x(s)| ds$$

$$\leq |x_0| + Nw + M \int_0^t |x(s)| ds$$

Aplicando lema Gronwall con $a = |x_0| + Nw$ $b = M$,

$$|x(t)| \leq a e^{b(t-0)} \leq a e^{bw} \rightarrow \lim_{t \rightarrow +\infty} |x(t)| < \infty \quad !!!$$

Por tanto, $w = +\infty$.



Ejemplo de uso de Tª Global: Ver que la solución de

$$\dot{x} = \frac{3}{x} + \frac{\sin t}{x^3} + 7x, \quad x(0) = 2 \quad \text{está definida en } \infty$$

$$\text{Sea } \mathbb{X} : D = \mathbb{R} \times \mathbb{R}^+ \rightarrow \mathbb{R} / \mathbb{X}(t, x) = \frac{3}{x} + \frac{\sin t}{x^3} + 7x \Rightarrow \mathbb{X} \in C^0$$

$$\mathbb{X}(t, 1) = 3 + \sin t + 7 \geq 9$$

Sea $I =]t_1, w[$ el intervalo maximal de la sol.

Supongamos $w < +\infty \Rightarrow$ Por Tª Prolongación

$$1) \lim_{t \rightarrow w^-} |x(t)| = +\infty \quad \vee \quad 2) \exists t_n \nearrow w / x(t_n) \rightarrow \{ (w, \eta) \in \partial D$$

- Veamos 2) no se cumple. Supongamos que a'

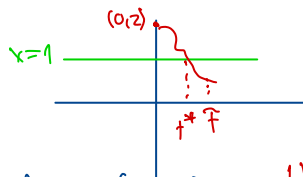
Veamos $x(t) > 1 \quad \forall t \in [0, w[$. Sup. que no lo es $\Rightarrow$

$$\exists \tilde{t} \in [0, w[/ x(\tilde{t}) \leq 1 \Rightarrow$$

$$\text{Sea } t^* = \min \{ t \in [0, \tilde{t}] / x(t) = 1 \}$$

$$x(t^*) = 1 \quad \text{y} \quad x(t) > 1 \quad \forall t \in [0, t^*[$$

$$\text{se tiene } 0 \leq \dot{x}(t^*) = \mathbb{X}(t^*, x(t^*)) = \mathbb{X}(t^*, 1) > 0 !!!$$



$$\partial D = \mathbb{R} \times \{0\} \Rightarrow \dot{x} = 0 \Rightarrow x(t_n) \searrow 0 !!! \text{ Imposible,}$$

pues $x(t) > 1 \quad \forall t \in [0, w[$

- Veamos 1) tampoco se cumple.

$$|x(t)| \leq |x(0)| + \int_0^t |\mathbb{X}(s, x(s))| ds \leq$$

$$2 + \int_0^t \frac{3}{|x(s)|} + \frac{1}{|x(s)|^3} + 7|x(s)| ds \leq 2 + 3 + t + 7 \int_0^t |x(s)| ds$$

$$\leq 2 + 4w + 7 \int_0^t |x(s)| ds. \quad \text{Por Lema Gronwall en } [0, w[$$

$$|x(t)| \leq a e^{bt} \leq a e^{bw}, \quad a = 2 + 4w, \quad b = 7 \quad \forall t \in [0, w[$$

$$\Rightarrow \lim_{t \rightarrow w^-} |x(t)| < +\infty !!! \Rightarrow w = +\infty$$

Exercício de Jorge:

Sea $g: \text{Iaib}[x \in \mathbb{C}id] \rightarrow \mathbb{R}$. Probar que son equivalentes:

$$(t, \lambda) \mapsto g(t, \lambda)$$

$$1) \lim_{t \rightarrow a^+} g(t, \lambda) = 0 \text{ u.i.g. } \forall \lambda \in [\text{cid}]$$

$$2) \forall t_1, t_2 \geq a, \quad g(t_n, \lambda) \rightarrow 0 \text{ u.i.g. } \forall \lambda \in [\text{cid}].$$

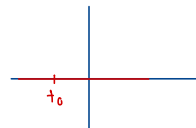
Sea el (P) $\begin{cases} \dot{x} = F(t, x) \\ x(t_0) = x_0 \end{cases}$. Probar que hay unicidad $\Leftrightarrow$ la sol. maximal es única.

Sea I intervalo acotado, $f_n: I \rightarrow \mathbb{R}^d / \{f_n\}$ univ. acotada y equicontinua,
 $\{f_n\} \rightarrow f: I \rightarrow \mathbb{R}^d$ puntualmente. Probar $\{f_n\} \xrightarrow{un} f$

Calcular la sol. general de $\dot{x} = x^2$ $x(t_0) = 0$

$$D = \mathbb{R} \times \mathbb{R}$$

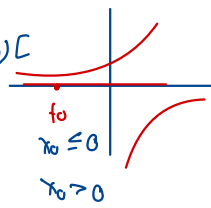
$$x(t_1, t_0, 0) = 0, \quad \forall t \in \mathbb{R}$$



$$x(t_1, t_0, x_0) \neq 0 \quad \forall t \in]\alpha(t_0, x_0), \omega(t_0, x_0)[$$

$$\int \frac{dx}{x^2} = \int dt \Rightarrow -\frac{1}{x} = t + C \Rightarrow C = -\frac{1}{x_0} - t_0 \quad x_0 \neq 0$$

$$x(t_1, t_0, x_0) = \frac{-1}{t + \frac{1}{x_0} - t_0} = \frac{x_0}{x_0(t_0 - t) + 1} \quad \forall t \in]\alpha(t_0, x_0), \omega(t_0, x_0)[$$



$$\alpha(t_0, x_0) = \begin{cases} -\infty & x_0 \geq 0 \\ t_0 + \frac{1}{x_0} & x_0 < 0 \end{cases}$$

$$\omega(t_0, x_0) = \begin{cases} +\infty & x_0 \leq 0 \\ t_0 + \frac{1}{x_0} & x_0 > 0 \end{cases}$$

Sea $g:]a, b[\times]c, d[\rightarrow \mathbb{R}$. Son equivalentes:

$$(t, \lambda) \mapsto g(t, \lambda)$$

$$1) g(t, \lambda) \xrightarrow{c.u.} 0 \quad t \rightarrow a^+, \quad \lambda \in]c, d[\quad 2) \forall t_n \searrow a, \quad g(t_n, \lambda) \xrightarrow{c.u.} 0 \quad \lambda \in]c, d[$$

1) $\Rightarrow$ 2)

$$\forall \varepsilon > 0, \exists \delta > 0 / 0 < t - a < \delta \Rightarrow |g(t, \lambda)| < \varepsilon \quad \forall \lambda \in]c, d[$$

$$t_n \searrow a, \exists N \in \mathbb{N} / \forall n \geq N \Rightarrow t_n - a < \delta \Rightarrow |g(t_n, \lambda)| < \varepsilon \quad \forall \lambda \in]c, d[$$

2) $\Rightarrow$ 1)

Veamos por contrarecíproca.

$$\neg 1) \Rightarrow \exists \varepsilon > 0 / \forall \delta > 0 \left. \begin{array}{l} \exists t, 0 < t - a < \delta \\ \exists \lambda \in]c, d[\end{array} \right\} / |g(t, \lambda)| \geq \varepsilon$$

$$\text{sea } n \in \mathbb{N}, \delta = \frac{1}{n} \Rightarrow \left. \begin{array}{l} \exists t_n, 0 < t_n - a < \delta = \frac{1}{n} \\ \exists \lambda_n \in]c, d[\end{array} \right\} / |g(t_n, \lambda_n)| \geq \varepsilon$$

$$t_n \searrow a, \sup_{\lambda \in]c, d[} |g(t_n, \lambda)| \geq |g(t_n, \lambda_n)| \geq \varepsilon$$

Sea la ecuación de Duffing (*) $\ddot{x} + x + \varepsilon x^3 = 0$. Probar que todas las soluciones están definidas en $\mathbb{R}$ si $\varepsilon > 0$.

$$\begin{array}{c} \rightarrow \rightarrow \leftarrow \leftarrow \\ \hline x=0 \quad \varepsilon > 0 \end{array} \qquad \begin{array}{c} \leftarrow \rightarrow \leftarrow \rightarrow \\ \hline x=0 \quad \varepsilon < 0 \end{array}$$

Suponemos masa $m=1$. Sea $I =]-\infty, \infty[$ el intervalo maximal.

$$E_{kin} = \frac{1}{2} \dot{x}^2, \quad E_{pot} = \int (x + \varepsilon x^3) dx = \frac{x^2}{2} + \frac{\varepsilon}{4} x^4$$

$$E = E_{kin} + E_{pot} = \frac{1}{2} \dot{x}^2 + \frac{x^2}{2} + \frac{\varepsilon}{4} x^4$$

Lema: Sea $x(t)$ una sol. maximal de (*) $\Rightarrow E(x(t))$ cte. $\forall t \in I$

Dem.:

$$\frac{d}{dt}(E(t)) = \dot{x}(t) \cdot \ddot{x}(t) + x(t) \dot{x}(t) + \varepsilon x(t)^3 \cdot \dot{x}(t) = \dot{x}(t) \underbrace{(\ddot{x} + x + \varepsilon x^3)}_0 = 0$$

$\Rightarrow E(x(t))$ cte. en I .

$$\text{Sea } \gamma = \begin{pmatrix} x \\ \dot{x} \end{pmatrix}, \quad X(t, x) = \begin{pmatrix} \gamma_2 \\ -\gamma_1 - \varepsilon \gamma_1^3 \end{pmatrix}, \quad X: D = \mathbb{R} \times \mathbb{R}^2 \rightarrow \mathbb{R}^2, \quad X \in C^1(D)$$

$$\partial D = \emptyset$$

$$\text{Por } T^a \text{ Unicidad, si } w < +\infty \Rightarrow \|\gamma(t)\| = \sqrt{x(t)^2 + \dot{x}(t)^2} \xrightarrow{t \rightarrow w^-} +\infty$$

$$\text{Sin embargo, } E(t) \geq \|\gamma(t)\|^2 \quad !!! \text{ Contradicción } \Rightarrow w = +\infty$$

$\downarrow$
cte.

Sea $\dot{x} = \Sigma(t, x)$, $x(t_0) = x_0$. Probar la existencia de $\frac{\partial x}{\partial t_0}(t; t_0, x_0)$

$$\Delta_h(t) = \frac{1}{h} [x(t; t_0+h, x_0) - x(t; t_0, x_0)] \in \mathbb{R}^d, h \neq 0, \text{ bien def.}$$

Si $t \in [a, b] \subseteq]\varphi(t_0, x_0), \omega(t_0, x_0)[$, $t_0 \in]a, b[$

$$\begin{aligned} \dot{\Delta}_h(t) &= \frac{1}{h} [\dot{x}(t; t_0+h, x_0) - \dot{x}(t; t_0, x_0)] = \frac{1}{h} [\Sigma(t; x(t; t_0+h, x_0)) - \Sigma(t; x(t; t_0, x_0))] \\ &= \frac{1}{h} \left[\int_0^1 \frac{d}{d\lambda} \Sigma(t, \lambda x(t; t_0+h, x_0) + (1-\lambda)x(t; t_0, x_0)) d\lambda \right] \end{aligned}$$

